

Debugging and Diagnosing

- Training to ~~not~~ like chiz maydi. Lekin yanov natija beradi. Shu holatda "naga model o'rganmayabdi" degan muammoni tekshiramiz.
- Training dynamics ni tahlil masak learning rate uchun "randomly" tanlashni alovatiramiz va holatni.

Train model $\Rightarrow$ yanov natija $\Rightarrow$ loss, tahlil qilamiz.

Aslida:

Observe $\Rightarrow$ Diagnose $\Rightarrow$ Apply correct fix.

- Yani:
- Muammoni topib uni ustidagi ilklarni
 - Unga diagnose (tahlil) qilyamiz.
 - To'g'ri yechimni beramiz.

- Masalan:
- Doktorga bordik.
 - Oxyrimiz sifriyani o'qitamiz. (muammo)
 - Doktor tahlil qilyadi (diagnose)
 - Dori yoki schedule tuzib b-oli. (debugging)

Diagnose $\Rightarrow$ Muammoni aniqlab unga g'anday yechim berish kerakligini topish.

Debugging $\Rightarrow$ ushbu yechimni "apply" qilish.

Underfitting
Overfitting
Oscillation
Exploding
Slow learning

uñur sabablar tufayli
model "diyaor" muhim.

Underfitting

Problem: $\Rightarrow$ Train loss - High
Validation loss - High

Diagnose: $\Rightarrow$ Yegoridagi holatda model
"Underfitting" bo'ladi. Buni hol
zotli uchun "Trainni ko'proq"
qilish k-k (Bigger model/train longer).

Debugging: $\Rightarrow$ Analize qilaish.

$\Rightarrow$ Train loss - Low
Val loss - High } overfitting } Regularization
Dropout/L2/
Early stopping

$\Rightarrow$ Train loss - Unstable
Val loss - Unstable } Oscillation } LR too big } Reduce LR

$\Rightarrow$ Train loss - Very high
Val loss - High } Exploding } Gradient explosion } lower LR / Batch Norm
Trainingni stabilizatsiya

$\Rightarrow$ Train loss - slowly $\downarrow$
Val loss - slowly $\downarrow$ } Slow learning } LR too small } Increase LR

